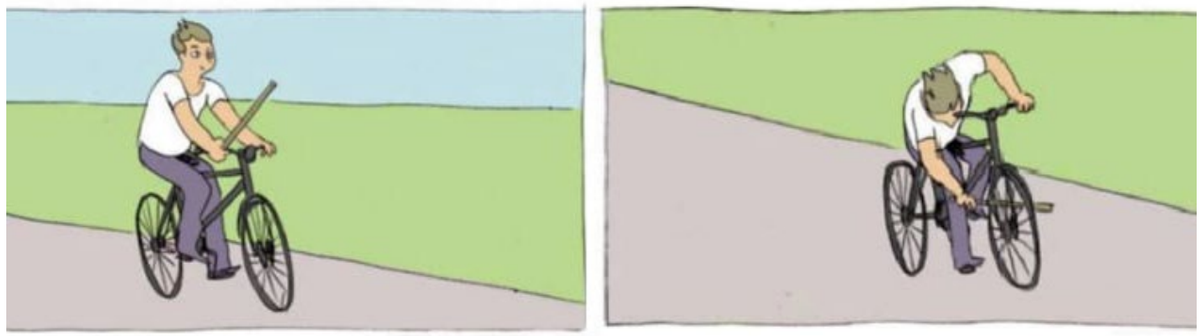
Object Oriented Programming 101

Object Oriented Concepts



- ☐ Encapsulation
- ☐ Inheritance
- ☐ Polymorphism
- ☐ Coupling
- ☐ Cohesion

Encapsulation

Encapsulation is the inclusion within a program object of all the resources needed for the object to function - basically, the method and the data. The object is said to "publish its interfaces." Other objects adhere to these interfaces to use the object without having to be concerned with how the object accomplishes it

Protection

Information Hiding

example

users should not have direct access to variables or “helper” methods

```
public class mammal {
        private int numberoflegs=4;

        int getlegs() {
                return numberoflegs;
        }

        private void makeDBconnection(){

        }
}
```

use getter & setter

Benefits

- maintainability, flexibility and extensibility: modify our implemented code without breaking the code of others who use our code

Inheritance

Inheritance is the concept that when a class of objects is defined, any subclass that is defined can inherit the definitions of one or more general classes

Example

mammal

```
public class mammal {
        int numberoflegs;
   int makeSound() { }
}
```

properties + methods

use this arrow

```
public class cat extends mammal { }
```

```
.....
cat kitty = new cat();
kitty.numberoflegs=4;
kitty.makeSound();
.....
```

inherit from mammal

Benefits

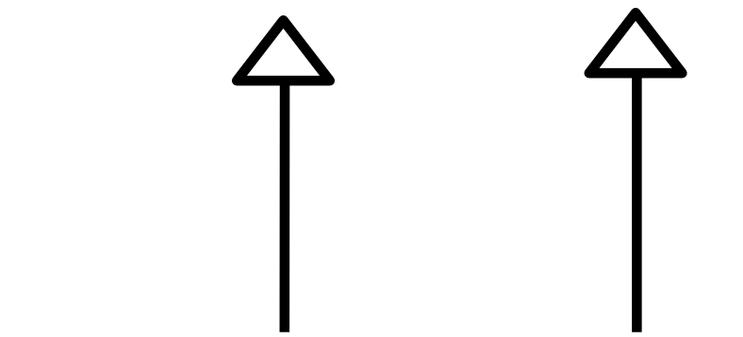
- Object in a subclass need not carry its own definition of data and methods that are generic to the class
  - »Maintenance: changes only have to be made in the superclass
  - »Reliability: what works for the superclass must work for the subclass.

Polymorphism

Polymorphism (having multiple forms) - Being able to assign a different meaning to a particular symbol or "operator" in different contexts

Example

mammal



```
mammal catwoman;
catwoman = new cat();
catwoman = new human();
```

can be 2 different objects

```
public class mammal {
        int numberoflegs=4;
}


public class human extends mammal {
        this.numberoflegs=2;
    void makeSound() {.”hello”... }
}
```

override parameter or method

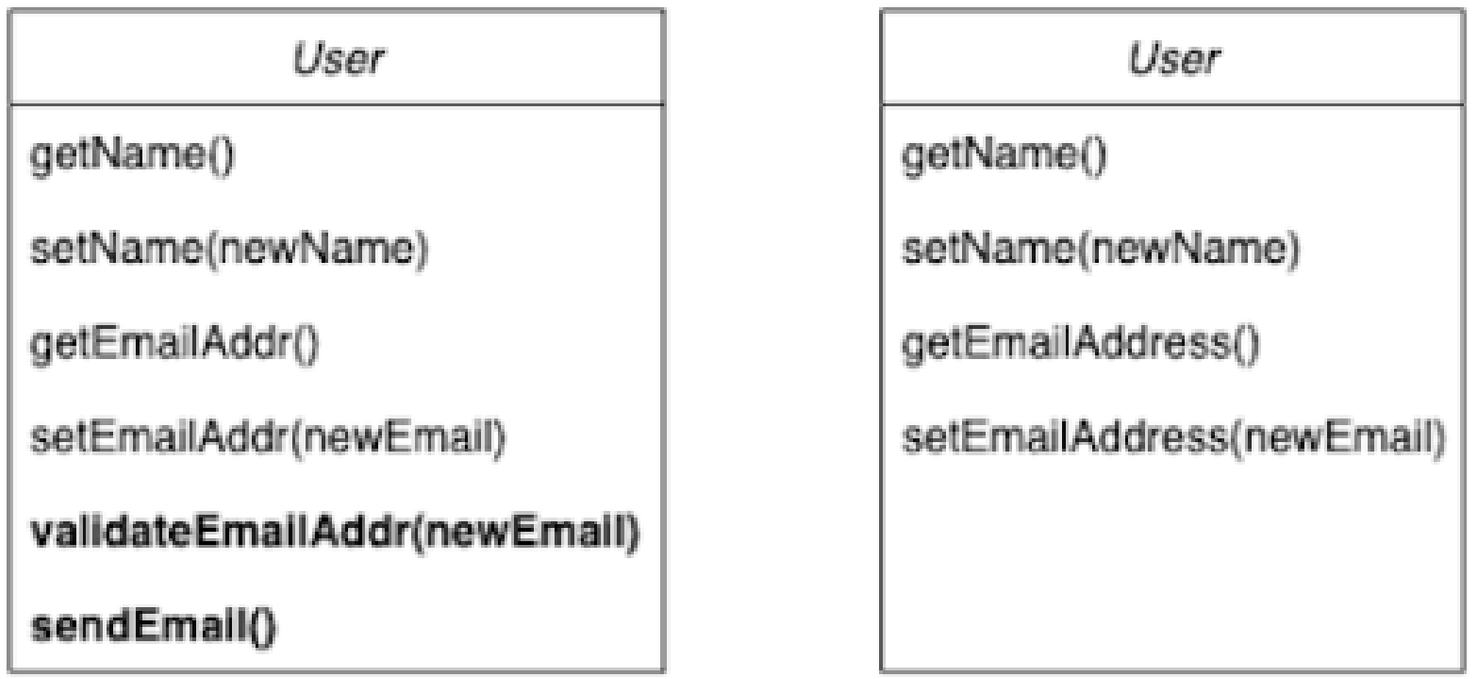
Benefits

- Variability: any object is able to respond differently to some common method or property
- Maintenance: objects are independent of each other.

Cohesion

- The degree to which the elements inside a module (or class) belong together
- Higher is better

Example



Low Cohesion

High Cohesion

Benefits

- Readability: increases ease of understanding
- Reusability: increases use of existing code
- Complexity: reduces complexity to a more manageable level

Coupling

- The degree of interdependency between modules.
- Tightly coupled modules must change together
- Looser is better

Example

Tight	**Loose**
If you want to change your skin, you would also have to change the design of your body as well because the two are joined together – they are tightly coupled.	If you change your shirt, then you are not forced to change your body – when you can do that, then you have loose coupling.

Benefits

- Replacing: increases ease of replacing or swapping code/components
- Testability: easier to test units in isolation
- Extendability: reduces risk of breaking a system when adding new features

Design Patterns

Design Patterns are "descriptions of communicating objects and classes that are customized to solve a general design problem in a particular context." -- Gamma et al.

Benefits

- ☐Help us solve recurring problems!
- ☐Provide us with common vocabulary.
- ☐Improve readability and maintainability.

UML 101

https://www.visual-paradigm.com/guide/uml-unified-modeling-language/uml-class-diagram-tutorial/

Java 101

https://www.tutorialspoint.com/java/index.htm
https://www.youtube.com/watch?v=eIrMbAQSU34

projector

desk

Group 1 Group 2 Group 3

Group 4 Group 5 Group 6

Group 7 Group 8 Group 9

Group 10 Group 11 Group 12

pillar

Group 13 Group 14

Group 15 Group 16 Group 17

Group 18 Group 19 Group 20

Group 21 Group 22 Group 23

Group 24 Group 25 Group 26

pillar

Group 27 Group 28 Group 29

skateboards

door